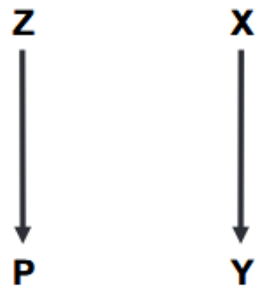


SURV/SURVMETH 720 —
Total Survey Error

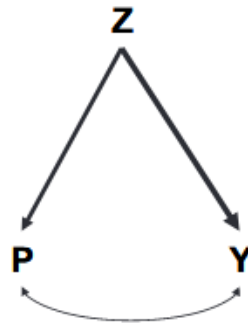
Dealing with Nonresponse

Class #5

September 29 2025



1. Separate Causes Model



2. Common Cause Model



3. Survey Variable Cause Model

Groves (2006)

Methods for Assessing Nonresponse Bias

Methods for Assessing Nonresponse Bias

1. Benchmarking (comparing *estimates*)
2. Comparing *characteristics* of respondents and non-respondents
3. Compare *responses* given by different *groups* of respondents (e.g., early vs late responders)
4. Compare *y* values (methodological studies)

1. Benchmarking

- Compare estimates obtained to estimates from other sources that are assumed to be more accurate
- Example: Keeter et al. (2000) compared estimates from “standard” 5-day survey (RR: 37%) to those from “rigorous” 8-week survey (RR: 61%)
 - Mean difference over 91 items around 2%
- Limitation: may confound NR error with other error sources

2. Compare Characteristics of Responders and Nonresponders

- Data about nonrespondents can come from:
 - Information on sampling frame
 - Paradata such as interviewer observations
 - Responses to screener interviews
 - Responses to other survey waves
 - Administrative records
- Example: Groves and Couper (1998)
 - Response to six household surveys evaluated using matched 1990 Census data
 - Lower contact rates for multiunit structures, single person households, households in urban areas.
 - Cooperation rates lower (refusal rates higher) in middle-aged households, those who live alone.

Table 8.2 A Comparison of Responding and Nonresponding 17-Year-Olds in the National Assessment of Education Progress^a

	Respondents	Nonrespondents
Auxiliary Variables		
Grade point average ^b		
Mean	2.6	2.2
Median	2.7	2.2
Number of Courses taken during most recent semester		
Mean	5.4	4.7
Median	5.5	5.2
Percentage of school days absent in current year		
Mean	5.9	11.3
Median	4.2	7.3

Lessler and Kalsbeek, p.172

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Percentage of school days absent in current year		
Mean	5.9	11.3
Median	4.2	7.3
Number of Correctly Answered Exercises^c		
Mathematics Package (total number of exercises)		
01 (16)	9.4	8.1
03 (19)	10.6	9.0
09 (19)	9.5	8.4
13 (22)	12.4	10.3
Science Packages		
01 (23)	12.0	10.7
03 (13)	6.2	5.6
09 (28)	18.9	17.9
13 (7)	3.2	2.9

3. Compare Responses Given by Different Groups of Respondents

- Early vs. late responders
 - May be informative about the direction of response bias
- Nonresponse follow-up studies
 - Goal: to obtain responses from a random sample of the nonrespondents
- Those who express willingness to participate in a future survey vs. those who do not

Curtin et al (2000)

- Background: Constant RR (~70%) but increased effort (mean #calls: ~ 4 to 8) needed over the years.
- What happens if we exclude respondents who
 - Needed refusal conversion ? (RR reduced by 5-10 %)
 - Needed more than 5 calls ? (RR reduces 25%)
 - Needed more than 2 calls ? (RR reduces 50%)

Small to no effect on key estimate! (and on the component estimates)

But is this really about absolute nonresponse bias ?

Index of Consumer Sentiment (y)

- While small,
 - Initial cooperators to Refusal conversions
 - y for Initial cooperators $>$ Initial refusers
 - Early call IWs to later call IWs
 - y for early calls $<$ later calls

4. Compare y values on respondents and non-respondents (methodological studies)

Amaya and Presser (2017)

- Use frame information from the American Time Use Survey (ATUS) and the Survey of Health, Ageing, and Retirement in Europe (SHARE).
 - ATUS drawn from CPS.
 - SHARE is a panel survey.
- Compare **estimates** from the full sample to the respondent sample.
- H1: Social integration hypothesis. Those who are social will also respond to surveys. So estimates on social activities and roles should be biased upwards. [yes]
- H2: Variables measuring political and social activities should be more biased than those measuring other activities (e.g., hosting a dinner party) [some evidence]
- H3: Regression coefficients predicting these activities unimpacted [impact present but small].

Reviewing: Relationship between response rates and nonresponse bias

Relationship between response rates and nonresponse bias

- Higher response rate decreases the risk of nonresponse bias, other things being equal. On average.
- But *overall* relationship is generally weak
 - Groves and Peytcheva, 2008 : 959 estimates from 59 studies: correlation between resp. rates and NR bias = 0.2 (R^2 of 0.04)
 - Brick and Tourangeau, 2017 : Study-level NR bias more correlated with the response rate than estimate-level bias ($R^2 = 0.23$)

Groves and Peytcheva (2006)

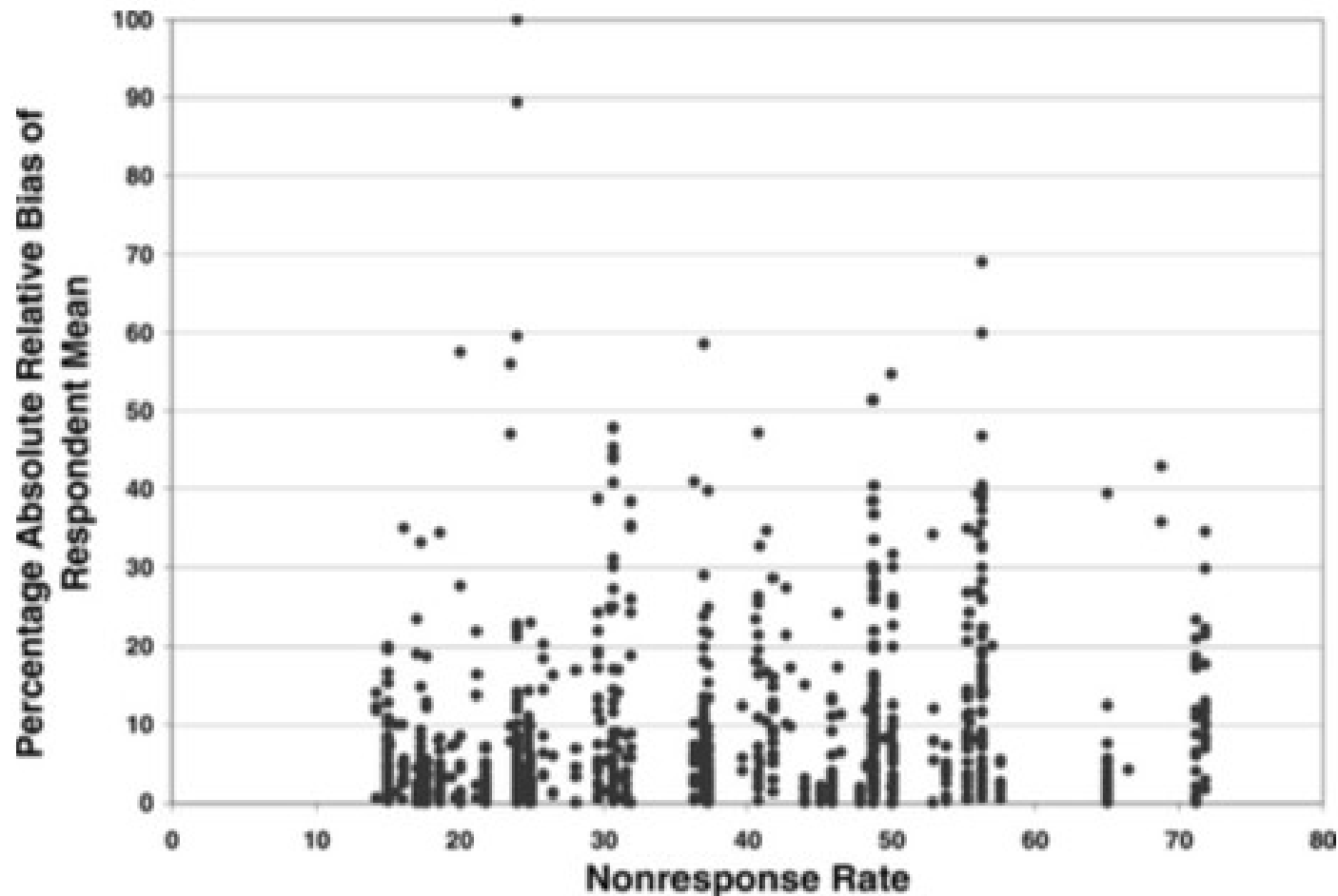
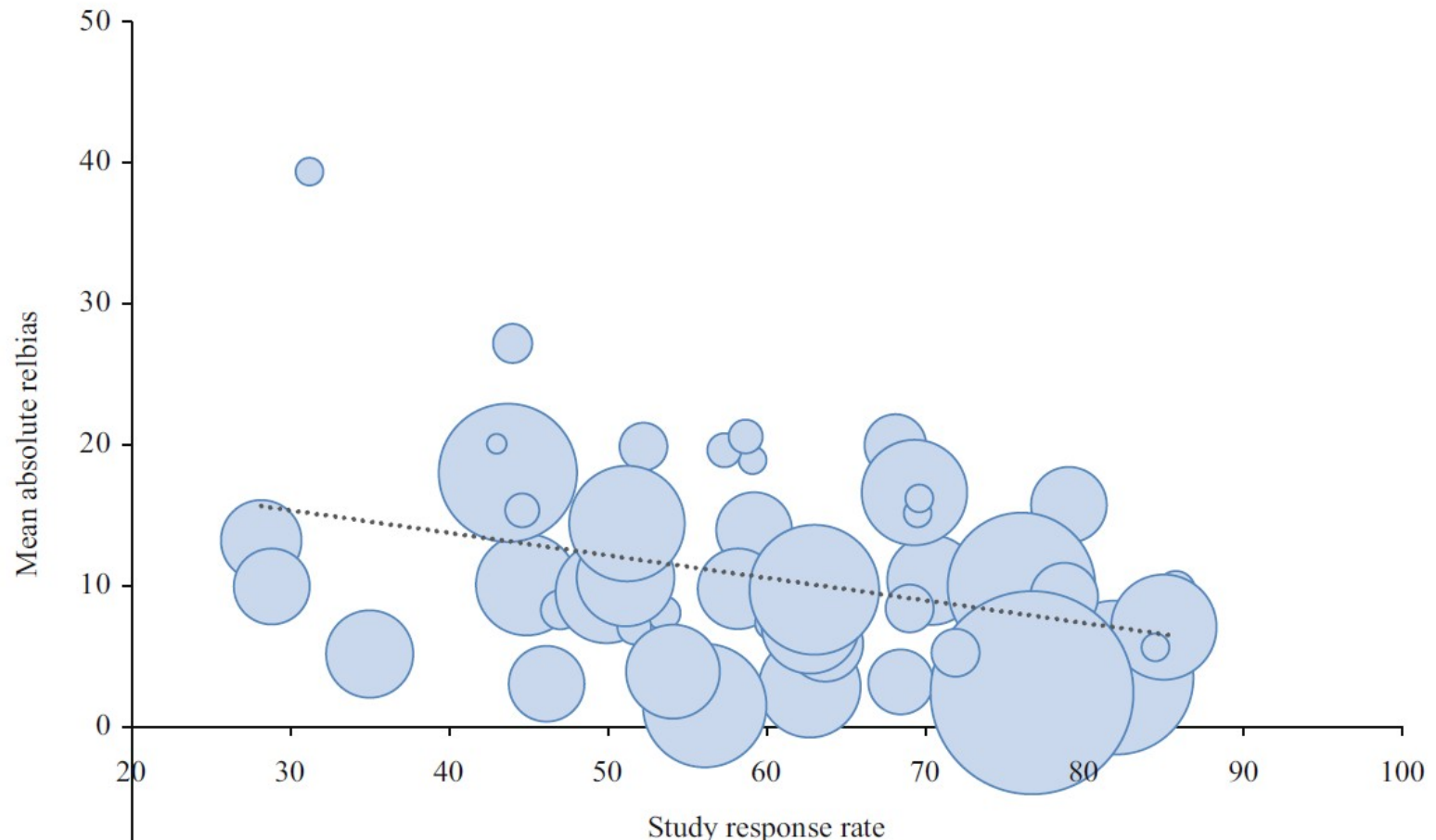


Figure 2. Percentage Absolute Relative Nonresponse Bias of 959 Respondent Means by Nonresponse Rate of the 59 Surveys in Which They Were Estimated.

Brick and Tourangeau (2017)



Bubble plots of the mean absolute relbias of the estimates against the study's response rate. The bubble sizes are proportional to the number of bias estimates from the study.

"Response rates may not be very good predictors of nonresponse bias, but they are far from irrelevant."

$$\text{Bias}(\bar{y}_r) = \frac{\sigma_y \rho}{\bar{\rho}} \approx cv(\rho) \sigma_y cor(y, \rho)$$

•Xinyu:

“In practice, should we be more concerned about a low average response rate, or about large variation in response propensities across individuals? And does the answer depend on the specific research question or variable of interest?”

Sagnik:

How realistic is it to assume that $cor(y, \rho)$ would remain stable—or at least not increase faster than the response rate—when responsive design interventions are applied?

Feiran:

While the random propensities model emphasizes transient, non-systematic influences on response, many real-world surveys observe consistent patterns of nonresponse across subgroups. How can we distinguish random nonresponse from systematic nonresponse, especially when the reasons for nonresponse are unobserved?

Can a higher response rate ever lead to
higher nonresponse bias?

Remember last week's question?

So, do you think you should prioritize achieving a high response rate?

Reasons not to blindly pursue a high response rate

- Weak empirical link between nonresponse rates and estimate-level bias
- Consumes resources that could be used to reduce other errors
- Focus could be on increasing response rates of groups that are underrepresented, not on increasing the overall response rate
- Focus could be on other metrics (e.g., "R-indicators") that are more closely related to nonresponse bias

Indicators of NR bias

Identification of NR Bias

- Groves et al. (2008) identifies two major types:
 - Survey level
 - Statistic level

A. Survey-level Indicators

- Variance of response rates on subgroups defined for all sample cases (respondents and nonrespondents)
 - As variance increases, presumed data quality decreases
- Goodness of fit statistics on propensity models
 - As model fit increases, presumed data quality decreases
- Variance of estimated response propensities (R-indicators)
 - As variance increases, presumed data quality decreases

Survey-level Indicators:

R-indicator

- Basic procedure:
 - Fit a regression model with “response” as dependent variable and unit characteristics as independent variables
 - Estimate the probability of response (p_i) for each responding unit using its characteristics
- Most basic R-indicator is $1 - 2 \text{SD}(\rho)$
 - ρ : response propensities; SD: standard deviation

R-indicator...

- As R-indicator increases, presumed data quality increases
- Points to NR bias if response probabilities are variable. Implies strong association between nonresponse and auxiliary variables
 - Maximum $S(\rho) = .5$ and R-indicator = 0
- Points to negligible NR bias if response probabilities are equal. Implies weak association between nonresponse and auxiliary variables.
 - In this case $S(\rho) = 0$ and R-indicator = 1
- Estimate ρ . Dependent on the model (comparing R-indicators for different surveys?)

Example: Dutch Labor Force Survey (LFS), 2005

- Two samples of non-respondents
 - Call Back: CAPI, full questionnaire
 - Basic-question: mixed-mode + shorter qnr.

Response	Response rate	R-indicator	Confidence interval
LFS	62.2%	0.80	(0.78 ; 0.83)
LFS + call-back	76.9%	0.85	(0.82 ; 0.88)
LFS + basic-question	75.6%	0.78	(0.76 ; 0.80)

<http://www.risq-project.eu/>
<http://hummedia.manchester.ac.uk/institutes/cmist/risq/R%20Indicators.pdf>

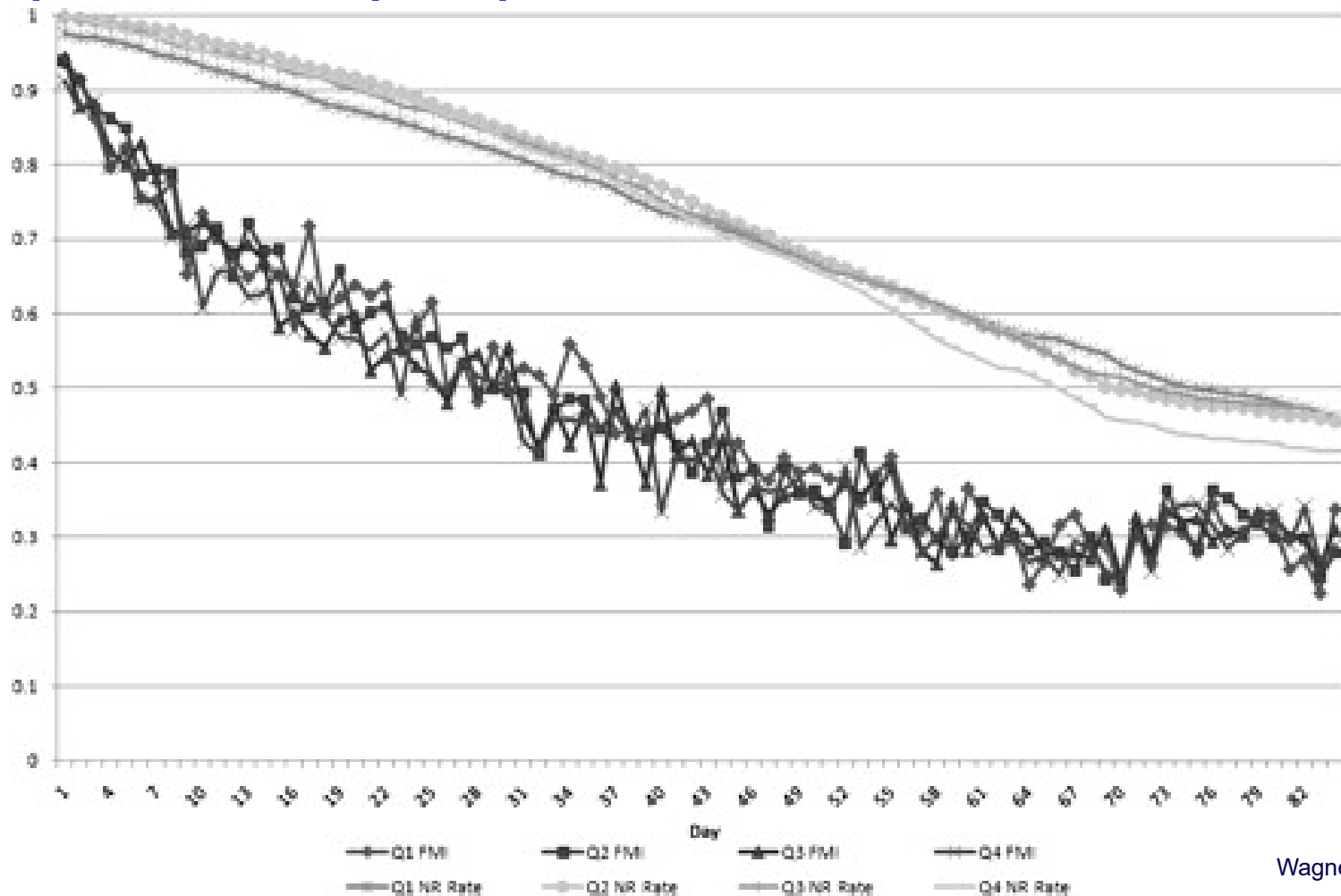
B. Estimate-level Indicators

- Comparisons of respondents and nonrespondents on auxiliary variables
 - As differences increase, presumed data quality decreases
- Correlation between post-survey nonresponse adjustment weights and y (from the respondent cases)
 - As correlation increases, presumed data quality decreases

Estimate-level indicators...

- Level of uncertainty about values one would impute for current nonresponders on y (“Fraction of Missing Information”)
 - Imputation models based on frame data and paradata
 - Missing data that cannot be recovered via nonresponse adjustments
 - As FMI decreases, presumed data quality increases

Example 1: NSFG - proportion of people who have never married

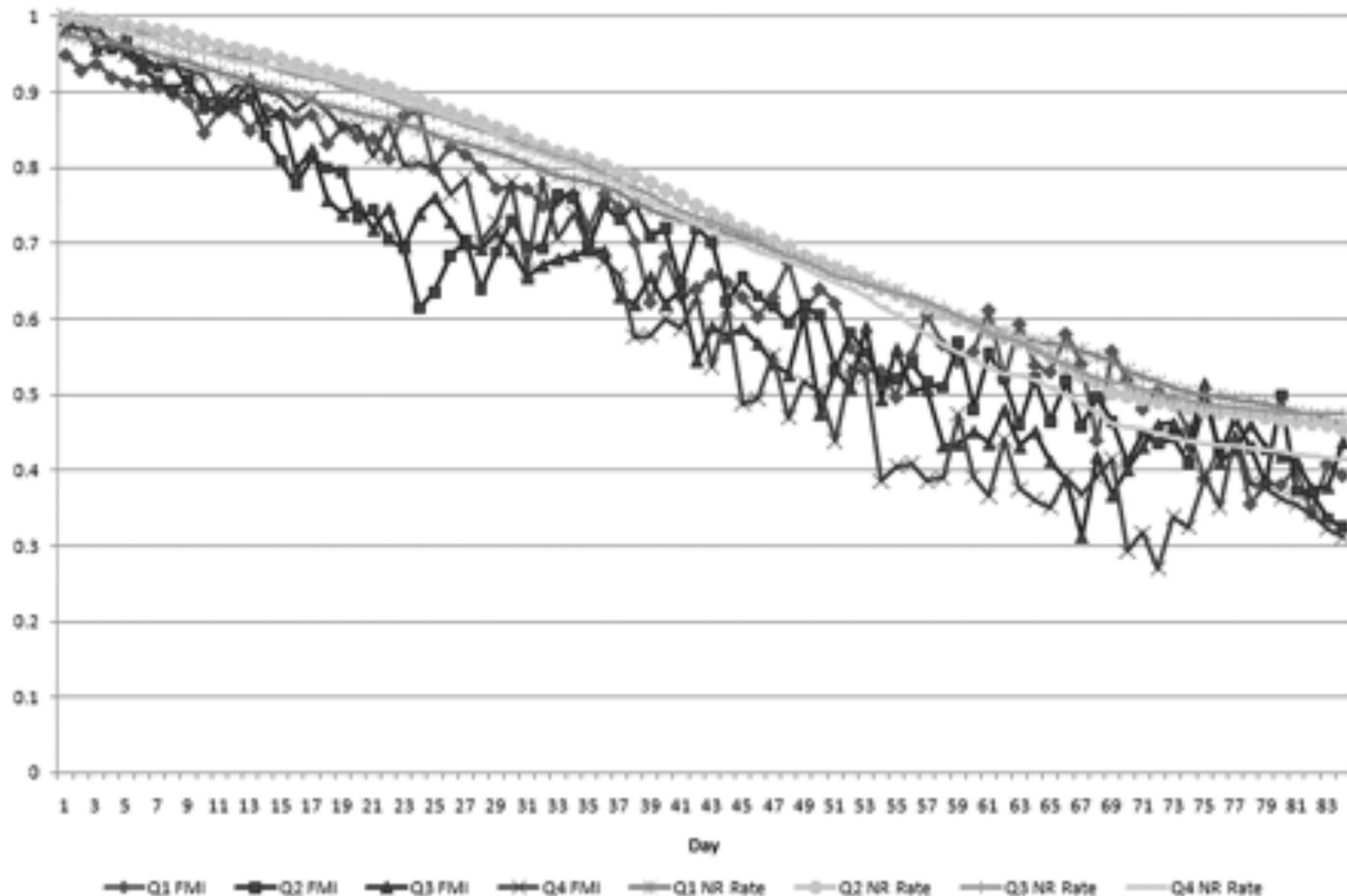


Wagner (2010)

Sample frame: ZIP-code level Census data for age; urbanicity

Paradata: contact attempts, lower observations (whether there are children under age of 15 in HH)

Example 2: NSFG – number of sexual partners in past 12 months



Summary on FMI

- Strengths:
 - Provides a sense of nonresponse bias after nonresponse adjustments have been made (other metrics only apply to unweighted sample)
- Weaknesses
 - Informative to the extent that similar adjustments are made to the final survey estimates
 - Cost (modeling and data inputs)

Dealing with non-response (reducing non-response bias)

1. Front-end techniques
2. Responsive and Adaptive designs

1. Front-end Techniques

1. Changing contact conditions

- Advance notification
- Use of priority mail
- Personalization of correspondence
- Showing respondent how data are used
- Establishing credibility

2. Follow-ups

- Repeated callbacks, visits
- Reminder cards, emails, or phone calls
- Non-response follow-up

Front-end Techniques ...

3. Changing respondent conditions
 - Reducing (perceived) burden of survey
 - Offering incentives for survey response
 - Allowing proxy respondents

4. Changing interviewer conditions
 - Interviewer characteristics
 - Interviewer training
 - Interviewer incentives

Front-end Techniques ...

- Strengths:
 - Firmly established in survey literature (LMICs ?)

- “Can Incentives Improve Survey Data Quality in Developing Countries?: Results from a Field Experiment in India” Stecklov et al, JRSS-A, 2018

Table 1. Household response statistics by city and treatment category†

	<i>Combined results</i>			<i>Results for Davanagare</i>		<i>Results for Gulberga</i>	
	<i>Overall</i>	<i>Treated</i>	<i>Control</i>	<i>Treated</i>	<i>Control</i>	<i>Treated</i>	<i>Control</i>
Total count	2333	1114	1219	539	639	575	580
Non-response count	50	1	49	0	47	1	2
Response rate	0.979	0.999	0.960	1.000	0.926	0.998	0.997
F-statistic	14.47 ($p = 0.000$)			16.37 ($p = 0.000$)		0.20 ($p = 0.656$)	

†Tests account for the stratified split-panel design by using Stata complex survey data methods.

- Contingent incentive (approx. equivalent to \$5)
- Our study suggests that incentives reduced unit non-response. Find little impact of incentives on a broad range of sociodemographic, behavioural and attitudinal questions.
- In contrast, we consistently find that households that received incentives reported substantially lower consumption and income levels and fewer assets. Given random assignment and very high response rates, the most plausible interpretation of this finding is that incentivizing respondents in this setting may increase their motivation to present themselves as more needy.

Front-end Techniques ...

- Strengths:
 - Firmly established in survey literature (not much for developing countries)
 - Can inform initial data collection protocol
- Limitations:
 - Cost
 - Lack of specific guidance
 - Focused on response rates, not bias

Dealing with non-response (reducing non-response bias)

1. Front-end techniques
2. Responsive and Adaptive designs

Responsive (and Adaptive) Survey Design

'Responsive' and 'adaptive' survey design often used interchangeably to describe a whole group of strategies for managing data collection including:

- front loading potentially difficult cases
- tailoring data collection strategies to different subgroups
- prioritizing effort according to estimated response propensities
- imposing stop rules for ending data collection
- monitoring key survey estimates throughout the field period
- using two-phase or multiphase sampling for following up non-respondents
- calculating indicators of non-response bias (such as the R-indicator) other than response rates to monitor and guide fieldwork.

(Tourangeau et al. 2017)⁴⁰

Responsive Survey Design

- Efforts to monitor data quality indicators during a survey in order to change design features on the basis of those indicators.
- Often involves concentrating resources on 'hard-to-interview' cases and reducing effort on 'easy' cases in the sample
 - Potential design changes to target 'hard' cases: more contacts, larger incentive, costlier mode, assigning most successful interviewers, etc.

While ‘adaptive’ and ‘responsive’ often used interchangeably:

- Responsive: “Phases”
- Multiple protocols introduced in a phased design over the course of a study.
- Concept of “phase capacity”:
 - Additional data not contributing to further changes in the survey estimates
 - Can be cost-inefficient compared to starting another phase with a different and complementary protocol

Jeremy:

- There is a large body of research identifying certain systematic response propensities, particularly for sensitive topics. ...significant documented bias due to survey sponsor, survey location (ie brief intervention in an emergency room or doctor's office), income, and cultural affiliation. These characteristics are extremely difficult to identify with follow-up with non-responders, though not impossible. Of the available main strategies (ie monetary, increased effort, mode switching), which are better indicators and how easily can this be validated?

- **Adaptive: “Tailoring”**
 - Some individuals within a phase can be assigned to a different treatment.
 - Different protocols can be applied to a subset of the sample, whether because they are more valuable for the survey estimates or because they are more likely to respond to the new protocol relative to the rest of the sample.
- Responsive and adaptive designs are complementary as adaptive designs in surveys are often implemented within phases of a responsive design.

Responsive/Adaptive Designs

- Divide field period into phases and let initial phase inform data collection protocols in later phase. *Examples:*
 - Phase 1 survey shows that single-person households have lower response rates than other households. In Phase 2 special efforts are made to recruit single-person households.
 - Phase 1 data indicated that, by 14 call attempts, the estimates were unlikely to significantly change. In Phase 2 call attempts are capped at 14.
- Change survey protocols during data collection.

Responsive/Adaptive Designs..

Three basic strategies:

- a) Multiphase designs
- b) Case prioritization
- c) Tailoring

Example of Case-prioritization

High School Longitudinal Study of 2009

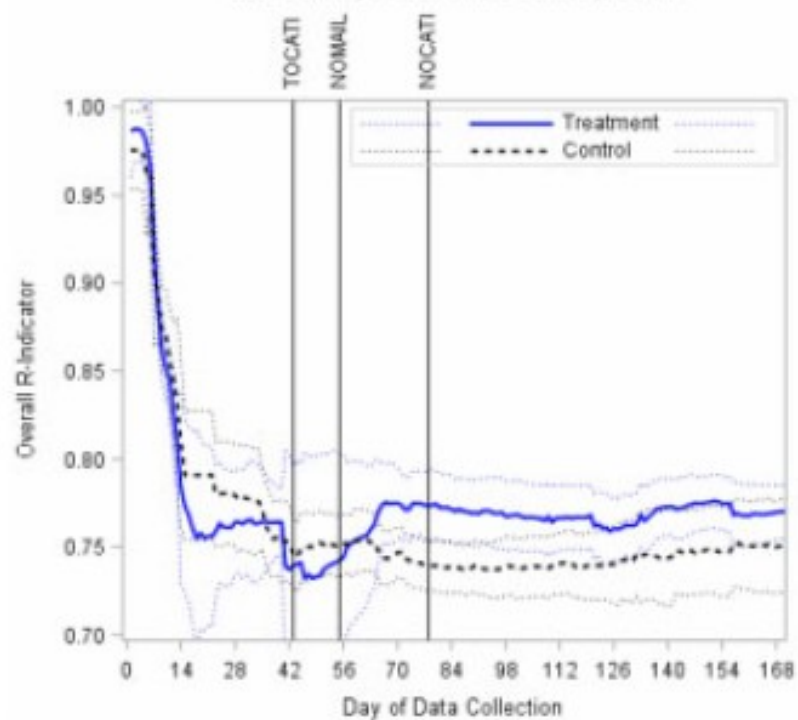
- Survey of parents; data collected initially by CATI and the Web
- Propensity models fit on six weeks' data.
- Pursue lowest propensity quartile cases using CAPI.
- Results:
 - Boosted response rate in the lowest propensity quartile
 - Significant bias reduction for some variables

Case prioritization especially useful when it impacts final estimates (rather than go by response propensities alone).

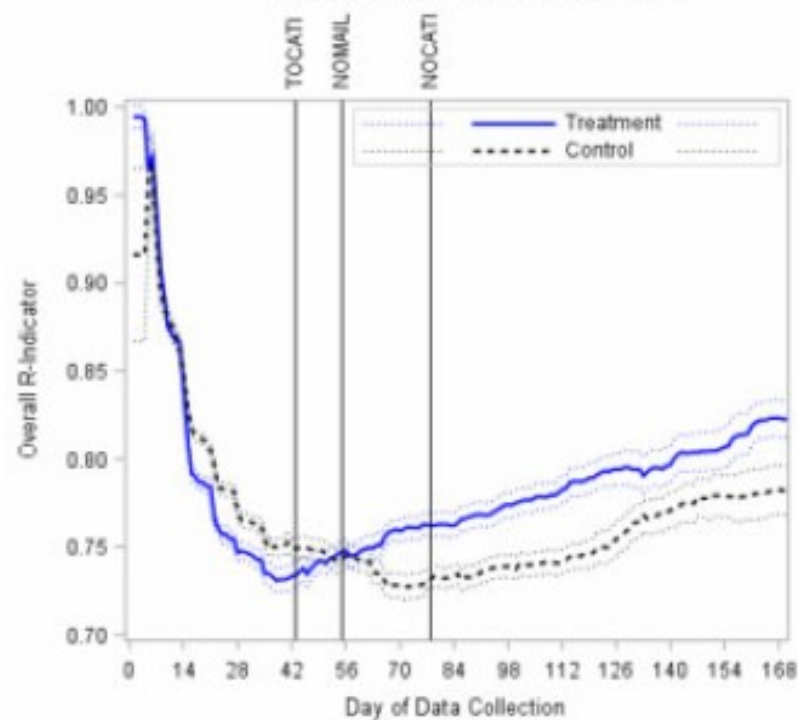
Example: Coffey et al. (2019) “*Dynamic adaptive design*”

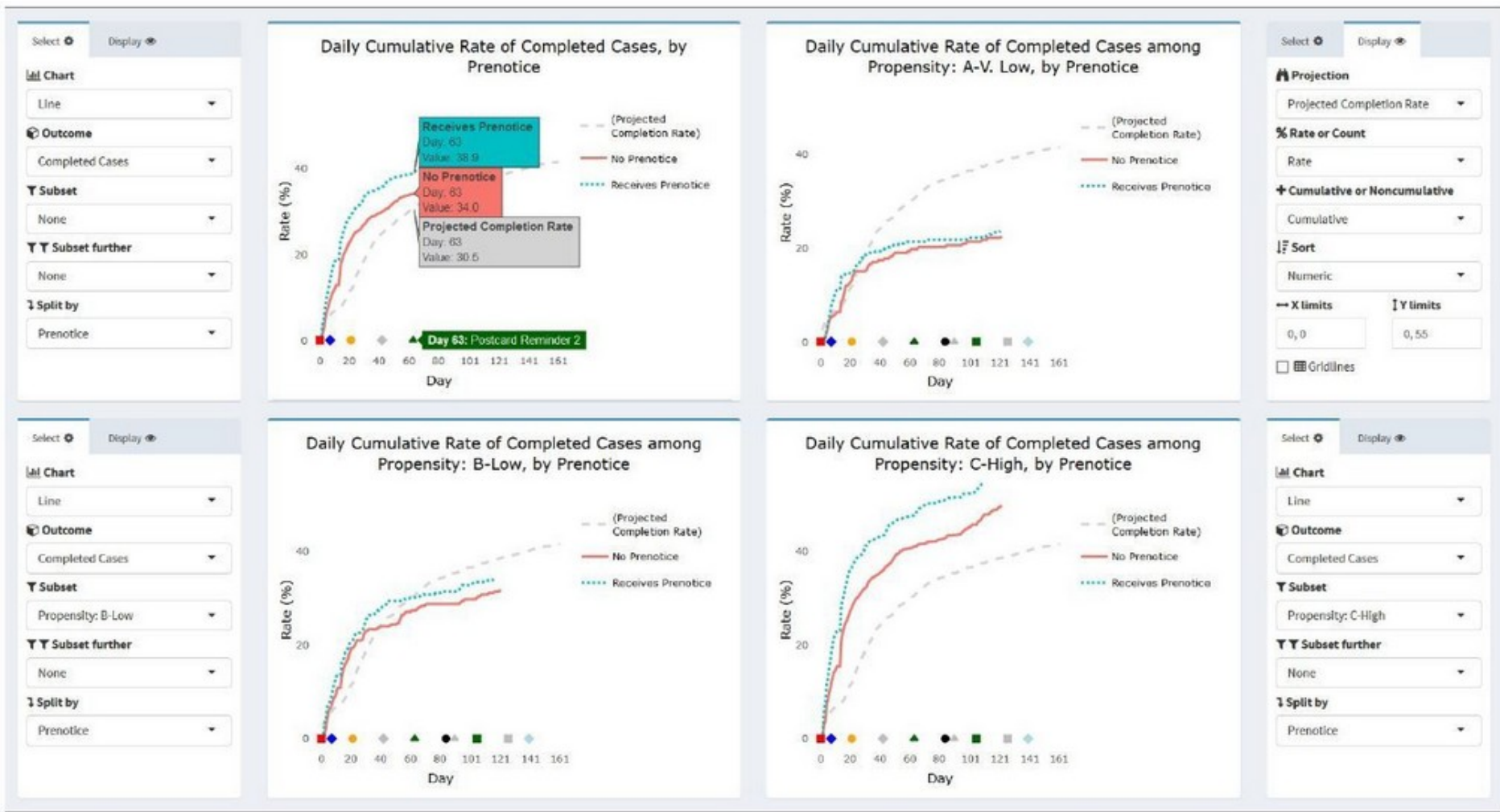
- National Survey of College Graduates (2015)
- Goal: Improve ‘representativeness’ while monitoring the effect of interventions on response rates and survey costs.
- Split-sample experiment: traditional methods vs DAD
- y: unemployment, income, etc.; Z: degree, gender, disability status, etc.
- Compute, closely watch, and intervene based on R-indicators.
 - Cases classified as ‘enhanced effort’, ‘reduced effort’, or ‘standard effort’
- Improved quality without increasing cost or reducing response rate

Overall R-indicators
New Cohort, Treatment vs Control, 95% CI



Overall R-indicators
Old Cohort, Treatment vs Control, 95% CI





Kehinde:

I am wondering if there are situations where the responsive design used in a survey is not effective at addressing certain large biases. In such cases, how do survey researchers identify and correct those remaining biases? And if they are unable to fix them, what happens to the data that has already been collected?

Responsive/Adaptive Design

- Strengths:
 - Can address sample balance not just response rates
 - Cost reduction
- Limitations:
 - Harder to implement because different survey protocols used for different groups
 - Can result in smaller sample size
 - Need good data on Rs and NRs to inform
 - Often requires metrics that are diagnostic of nonresponse bias that can be monitored during the study

Some more resources...

- Responsive survey design: Checklist for practitioners
- JSSAM Special Virtual Issue on Nonresponse Rates and Nonresponse Adjustments
- Interactive visualization to facilitate monitoring longitudinal survey data and paradata

Sau: how is unequal effort done in practice ?

Next Week

- Post-survey weighting
- Reading assignment
- Questions?